

GARP Risk and AI Exam Prep

100 Practice Questions with Theory

Unofficial Study Aid

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Question

A bank uses AI models for detecting fraudulent transactions. While the models are highly effective, the risk team is concerned about inscrutability. Which statement correctly describes inscrutability?

- A. Inability to understand the internal mechanisms of a model.
- B. Inability to determine the accuracy of a model's prediction.
- C. Inability to explain a model's predictions.
- D. Inability to generalize a model to out-of-sample data.

Answer: A

Explanation

Inscrutability = inability to understand internal mechanisms. Distinct from accuracy, explainability, and generalizability.

Theory

Inscrutability = cannot understand internal mechanisms.

Explainability = can explain why a decision was made.

Interpretability = can understand internal workings.

Key points:

Deep neural nets are accurate but inscrutable.

Inscrutability is a model risk (blocks validation, audit, compliance).

GARP exam tests the difference between these three terms.

Question

Deep Blue defeated Kasparov in 1997. Which best describes a characteristic of classical AI?

- A. Classical AI optimizes using gradient descent.
- B. Classical AI cannot process time series data.
- C. Classical AI relies on pre-defined rules in simple contexts.
- D. Classical AI scales easily to complex problems.

Answer: C

Explanation

Classical AI (GOFAI) uses rules and logic. No gradient descent (that is deep learning). Does not scale to complex problems.

Theory

Classical AI (GOFAI):

- Rule-based, symbolic logic
- Search algorithms (A^* , minimax, lookahead)
- Simple, well-defined contexts (chess, tic-tac-toe)
- Cannot scale to complex problems

Deep Learning:

- Neural networks, many layers
- Backpropagation + gradient descent
- Self-training (learns by competing against itself)
- Handles unstructured data (images, text, audio)
- Inscrutable (black-box)

Exam tip: “Self-training” or “gradient descent” = deep learning.

Question

Which best describes a societal risk scenario regarding AI?

- A. Trading company errors cause client losses.
- B. Social media company spreads misinformation influencing public opinion.
- C. Manufacturer hiring decisions are poor.
- D. Hospital accesses personal info causing privacy loss.

Answer: B

Explanation

Societal risk = harm to society at large. Misinformation influencing public opinion is societal.

Theory

Three levels of AI risk:

Individual: harm to a person (privacy, wrong diagnosis).

Organizational: harm to a company (losses, reputation, fines).

Societal: harm to society (misinformation, inequality, trust erosion).

Exam tip: “Society”, “public opinion”, “democracy” = societal risk.

Question

Which is a key differentiating feature of deep learning over classical AI?

- A. A* algorithm
- B. Recursive adversarial search
- C. Lookahead strategy
- D. Self-training

Answer: D

Explanation

Self-training is a deep learning feature. A*, adversarial search, and lookahead are classical AI.

Theory

AI breakthroughs:

1950s–1980s: Classical AI (logic, search, expert systems)

1997: Deep Blue beats Kasparov

2010s: Deep learning revolution (AlexNet, AlphaGo)

2017: Transformers (“Attention Is All You Need”)

2020s: Large Language Models (GPT, Claude, Gemini)

Deep learning features:

Self-training, backpropagation, gradient descent

Automatic feature extraction

Handles unstructured data

Inscrutable (black-box)

Question

A survey asks customers to select investment types (stocks, bonds, ETFs). How should these results be classified?

- A. Ordinal data
- B. Nominal data
- C. Interval data
- D. Ratio data

Answer: B

Explanation

Categories with no order = nominal data.

Theory

Four levels of measurement:

Nominal: categories with no order (stocks, colors)

Ordinal: categories with order, no equal intervals (education, Likert)

Interval: equal intervals, no true zero (temperature, IQ)

Ratio: equal intervals, true zero (income, age, height)

Exam tip:

“Types of” = nominal

“Ranked” = ordinal

“Temperature/score” = interval

“Amount/count” = ratio

Question

Which statement is correct regarding data scaling?

- A. Standardization creates a distribution bounded by 0 and 1.
- B. Standardization transforms skewed data into standard normal.
- C. Normalization results in data bounded by -1 and 1.
- D. Normalization preserves correlations between data points.

Answer: D

Explanation

Standardization = mean 0, SD 1 (not bounded). Normalization = bounded 0–1. Normalization preserves correlations.

Theory

Standardization (Z-score):

$$z = (x - \mu) / \sigma$$

Mean = 0, SD = 1

Not bounded by 0 and 1

Does not change distribution shape

Normalization (Min-Max):

$$x' = (x - \min) / (\max - \min)$$

Bounded by 0 and 1

Preserves correlations

Sensitive to outliers

Exam tip: “Bounded 0–1” = normalization. “Mean 0, SD 1” = standardization.

Question

Based on the AIC table, which model should the analyst select?

Step	Model	AIC
0	No predictors	3202.69
1	Age	3193.36
1	Age2	3203.33
1	Age3	3204.66
1	Age4	3203.63
1	Age5	3202.20
2	Age + Age2	3125.50
2	Age + Age3	3123.20
2	Age + Age4	3126.69
2	Age + Age5	3132.87
3	Age + Age3 + Age2	3125.19
3	Age + Age3 + Age4	3124.77
3	Age + Age3 + Age5	3124.50

- A. Age + Age3
- B. Age3
- C. Age + Age5
- D. Age + Age3 + Age5

Answer: A

Theory

Stepwise regression:

Forward: add one at a time

Backward: remove one at a time

Bidirectional: both

AIC (Akaike Information Criterion):

Lower AIC = better model

$$AIC = 2k - 2\ln(L)$$

k = parameters, L = likelihood

Exam tip: If adding a variable increases AIC, stop.

Question

Which is a key characteristic of machine learning compared to classical statistics?

- A. ML focuses on inference and parameter estimation.
- B. ML emphasizes prediction accuracy over interpretability.
- C. ML requires strict assumptions about data distribution.
- D. ML is always more interpretable than classical statistics.

Answer: B

Explanation

ML prioritizes prediction. Classical statistics focuses on inference and assumptions.

Theory

Classical statistics:

- Goal: inference (understand relationships)
- Parameter estimation, confidence intervals
- Strict assumptions (normality, independence)
- More interpretable

Machine learning:

- Goal: prediction (accuracy on unseen data)
- Generalization over inference
- Fewer assumptions
- Often less interpretable (black-box)

Exam tip: “Inference” = classical stats. “Prediction” = ML.

Question

Features include age (20–80) and income (20,000–500,000). Which technique should be applied?

- A. Remove the income feature
- B. Apply feature scaling
- C. Add more features
- D. Reduce observations

Answer: B

Explanation

Feature scaling prevents large-scale features from dominating distance calculations.

Theory

Why scale?

Distance-based algorithms use magnitudes

Income (20k–500k) vs age (20–80) — income dominates

When to scale:

Distance-based: K-means, KNN, SVM, PCA

Gradient-based: neural networks, logistic regression

Not needed: decision trees, random forests

Methods: Standardization (mean 0, SD 1) or Normalization (0–1).

Question

What is the primary purpose of PCA?

- A. Increase number of features
- B. Reduce dimensionality while preserving variance
- C. Classify data
- D. Cluster data

Answer: B

Explanation

PCA reduces dimensionality by transforming features into uncorrelated components while preserving variance.

Theory

PCA:

- Dimensionality reduction technique
- Creates uncorrelated components
- Ordered by variance explained
- First component = most variance

Benefits:

- Reduces overfitting
- Speeds up training
- Handles multicollinearity
- Enables visualization

Limitations:

- Hard to interpret components
- Assumes linear relationships
- Sensitive to scaling

Exam tip: PCA is unsupervised.

Question

How would you choose the number of clusters in unsupervised learning?

- A. Always use $K = 2$
- B. Use a scree plot and look for the elbow
- C. Set K equal to number of data points
- D. Use highest possible K

Answer: B

Explanation

Scree plot: chart WCSS vs K. Elbow = optimal K. Silhouette scores can also be used.

Theory

Scree plot (elbow method):

Plot WCSS (inertia) vs K

Look for elbow where decrease sharply changes

Beyond elbow, diminishing returns

Silhouette score:

$$s = (b - a) / \max(a, b)$$

a = avg distance to own cluster

b = avg distance to nearest other cluster

Range -1 to 1 ; higher = better

Rule of thumb: $K \approx \sqrt{N/2}$

Question

What are the steps in the K-means clustering algorithm?

- A. Specify K, scale, select centroids, assign, recalculate, repeat
- B. Specify K, assign labels, train classifier, evaluate
- C. Select K points, remove outliers, apply PCA, cluster
- D. Normalize, hierarchical cluster, cut dendrogram

Answer: A

Explanation

K-means: (1) Specify K, (2) Scale features, (3) Random centroids, (4) Assign points, (5) Recalculate centroids, (6) Repeat until stable.

Question

How would you choose between clusters from different initial centroid choices?

- A. Highest WCSS
- B. Lowest WCSS
- C. Most clusters
- D. First solution found

Answer: B

Explanation

Lowest WCSS = best fit to feature data.

Theory

WCSS (Within-Cluster Sum of Squares):

- Measures compactness
- Sum of squared distances to centroid
- Lower = better
- Always decreases as K increases

BCSS (Between-Cluster Sum of Squares):

- Measures separation
- Sum of squared distances between cluster centroids and overall centroid
- Higher = better separation

Exam tip: WCSS = compactness, BCSS = separation.

Question

How do you use a scree plot for K-means?

- A. Highest point
- B. Lowest point
- C. Elbow where gradient changes from steep to flat
- D. Point where WCSS equals zero

Answer: C

Explanation

The elbow = optimal K. Beyond it, adding clusters gives diminishing returns.

Theory

Scree plot:

X-axis: K

Y-axis: WCSS

Always decreases

Look for elbow (max curvature)

Elbow:

Before: steep decrease

After: flat decrease

Optimal trade-off

Pitfalls:

Ambiguous elbow

No visible elbow

Use domain knowledge + silhouette

Question

What are the two types of hierarchical clustering and their advantages?

- A. Divisive and agglomerative; no need to pre-define K, dendrograms interpretable
- B. Supervised and unsupervised; faster, more accurate
- C. Partitional and density-based; handles outliers
- D. Single and complete linkage; simpler, more scalable

Answer: A

Explanation

Divisive (top-down) and agglomerative (bottom-up). No need to pre-define K. Dendrograms are interpretable.

Theory

Approaches:

Divisive: top-down splitting

Agglomerative: bottom-up merging

Linkage criteria:

Single: shortest distance

Complete: greatest distance

Average: average distance

Ward's: minimizes variance

Advantages:

No need to pre-define K

Dendrogram shows all clusterings

Uncovers nested relationships

Question

True or False: K-means with Euclidean distance can only be used when clusters are approximately spherical.

- A. True
- B. False
- C. True only for large datasets
- D. True only for small datasets

Answer: A

Explanation

True. K-means assumes spherical clusters. Manhattan distance produces rhombus-shaped clusters.

Theory

K-means produces spherical clusters:

Euclidean distance → spherical

Manhattan distance → rhombus

Problems:

Outliers distort centroids

Curse of dimensionality

Non-spherical clusters fail

Correlated features struggle

Solutions:

Standardize, remove outliers

Apply PCA

Use kernel functions or DBSCAN

Question

True or False: WCSS will never rise when K increases.

- A. True
- B. False
- C. True only if K is even
- D. True only if K is odd

Answer: A

Explanation

True. Adding clusters cannot increase WCSS. At $K = N$, $WCSS = 0$.

Theory

WCSS is monotonically non-increasing in K:

Adding cluster never increases WCSS

At $K = N$, $WCSS = 0$

But $K = N$ is overfitting

Analogy:

Regression: adding features never decreases R^2

K-means: adding clusters never increases WCSS

Implication:

Cannot choose K by WCSS alone

Use elbow, silhouette, or domain knowledge

Question

What is a dendrogram and how is it interpreted?

- A. Bar chart of feature importance
- B. Tree-like diagram; long vertical lines = important clusters
- C. Scatter plot of clusters
- D. Line chart of WCSS vs K

Answer: B

Explanation

Dendrogram = tree showing hierarchical clustering. Long vertical lines = important cluster boundaries.

Theory

Dendrogram:

X-axis: data points

Y-axis: distance

Each merge = horizontal line

Height = merge distance

Interpretation:

Long vertical lines = important clusters

Short lines = minor clusters

Cut horizontally to choose K

Number of vertical lines crossed = K

Q19 — Centroid Calculation

Question

Data on three features for banks A, B, C. What is the centroid?

Feature	A	B	C
Customers (M)	1.2	6.0	0.5
Loan book (bn)	5	25	7
Branches	80	400	50

- A. (2.567, 12.333, 176.667)
- B. (2.567, 12.333, 176.667)
- C. (2.567, 12.333, 176.667)
- D. (2.567, 12.333, 176.667)

Answer: A

Explanation

Average: $(1.2+6.0+0.5)/3 = 2.567$; $(5+25+7)/3 = 12.333$; $(80+400+50)/3 = 176.667$.

Theory

Centroid = mean of cluster points:

- Average each feature across all points
- Recalculated each K-means iteration
- Does not have to be an actual data point

Formula:

$$\mu_j = \frac{1}{n_k} \sum_{i \in C_k} x_i$$

Exam tip: Add and divide by count.

Question

Which issue might pose problems for K-means?

- A. Outliers
- B. High dimensionality
- C. Non-linearity
- D. Non-spherical clusters

Answer: D

Explanation

K-means fails with non-spherical clusters. It assumes spherical clusters.

Theory

Two pathological cases:

Concentric circles: centroids end up in same place

Elongated clusters: edge points misallocated

Why K-means fails:

Assumes spherical, similar-size clusters

Uses distance to centroid only

Solutions:

Kernel K-means

DBSCAN

Spectral clustering

Gaussian Mixture Models

Question

What is the main advantage of DBSCAN over K-means?

- A. DBSCAN is faster
- B. DBSCAN finds arbitrary-shaped clusters and handles outliers
- C. DBSCAN requires fewer parameters
- D. DBSCAN always produces spherical clusters

Answer: B

Explanation

DBSCAN finds arbitrary-shaped clusters and is robust to outliers (noise points).

Theory

DBSCAN:

Clusters = dense regions

Core point: at least minPts neighbors within radius ϵ

Border point: within ϵ of core, not core itself

Noise point: neither = outlier

Advantages:

Arbitrary shapes

Robust to outliers

No need for K

Disadvantages:

Varying densities

Sensitive to ϵ and minPts

Question

What is the purpose of the silhouette score?

- A. Measure accuracy
- B. Compare cluster cohesion and separation
- C. Count clusters
- D. Visualize high-dimensional data

Answer: B

Explanation

Compares distance to own cluster (cohesion) with distance to nearest other cluster (separation).

Theory

Silhouette score for point i :

$$s_i = \frac{b_i - a_i}{\max(b_i, a_i)}$$

a_i = mean distance to own cluster

b_i = mean distance to nearest other cluster

Range: -1 to 1

Interpretation:

$s \approx 1$: well inside cluster

$s \approx 0$: boundary

$s < 0$: possibly mis-clustered

Choose K with highest average silhouette.

Question

What is the curse of dimensionality in K-means?

- A. Algorithm too slow with many observations
- B. Data points become sparse in high dimensions
- C. Number of clusters increases exponentially
- D. Centroids become unstable

Answer: B

Explanation

As features increase, data becomes sparse. Distances become similar, making clustering hard.

Theory

Curse of dimensionality:

- Distances become similar with more features
- K-means cannot distinguish clusters
- Convergence slows

Solutions:

- Remove irrelevant features
- Apply PCA
- Use cosine distance
- Use algorithms designed for high dimensions

Question

What is the difference between single linkage and complete linkage?

- A. Single uses shortest distance; complete uses greatest
- B. Single uses greatest; complete uses shortest
- C. Single for agglomerative; complete for divisive
- D. Single for divisive; complete for agglomerative

Answer: A

Explanation

Single linkage = shortest distance between clusters. Complete = greatest distance.

Theory

Linkage = cluster distance measure:

Single: shortest distance between any two points

Complete: greatest distance

Average: average distance

Ward's: minimizes within-cluster variance

Properties:

Single: elongated clusters, sensitive to outliers

Complete: compact clusters, robust to outliers

Average: balance

Exam tip: Single = shortest. Complete = greatest.

Question

What is the purpose of K-means++ initialization?

- A. Speed up convergence by choosing centroids far apart
- B. Reduce number of clusters
- C. Increase WCSS
- D. Eliminate outliers

Answer: A

Explanation

K-means++ spreads initial centroids apart, reducing poor clustering and speeding convergence.

Question

Given X and W , what is the feature map F ?

$$X = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 0 \\ 3 & 2 & 2 \end{pmatrix}, \quad W = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}$$

- A. $F = \begin{pmatrix} 1 & -1 \\ 4 & 4 \end{pmatrix}$
- B. $F = \begin{pmatrix} 1 & 4 \\ -1 & 4 \end{pmatrix}$
- C. $F = \begin{pmatrix} 2 & 3 \\ 6 & 4 \end{pmatrix}$
- D. $F = \begin{pmatrix} 2 & 6 \\ 3 & 4 \end{pmatrix}$

Answer: A

Explanation

$F_{11} = 0(-1) + 1(0) + 1(1) + 0(1) = 1$; $F_{12} = -1$; $F_{21} = 4$; $F_{22} = 4$.

Theory

Convolution:

Slide kernel over input

Multiply overlapping, sum

Result = feature map

Example (2x2 kernel on 3x3):

Output size = $(3 - 2 + 1) \times (3 - 2 + 1) = 2 \times 2$

$$F_{11} = X_{11}W_{11} + X_{12}W_{12} + X_{21}W_{21} + X_{22}W_{22}$$

Used in: CNNs for image recognition.

Question

Analyst samples with replacement, creates multiple trees using all features, averages. Which technique?

- A. Adaptive boosting
- B. Random forest
- C. Gradient boosting
- D. Bootstrap aggregation

Answer: D

Explanation

Bagging uses all features. Random forest uses random subset of features.

Theory

Ensemble = combine models:

Bagging: bootstrap samples, all features, average

Random Forest: bagging + random feature subset

Boosting: sequential, correct errors

AdaBoost: reweights misclassified

Gradient Boosting: fits residuals

Exam tip:

“All features” + “average” = bagging

“Random subset” = random forest

“Sequential” = boosting

Question

What is an advantage of random forest?

- A. Captures previous misclassifications
- B. Improves interpretability
- C. Improves accuracy by reducing correlation across trees
- D. Captures outliers via Cook's distance

Answer: C

Explanation

Random forest reduces correlation across trees, improving prediction accuracy.

Theory

Random Forest:

- Each tree on bootstrap sample
- Random feature subset at each split
- Average (regression) or majority vote (classification)

Advantages:

- Reduces overfitting
- Reduces correlation across trees
- Handles high dimensions
- Robust to outliers
- Feature importance

Disadvantages: Less interpretable, expensive.

Question

Boundary: $0.43x_1 + 2.55x_2 + 0.67 = 0$. For $x_1 = 0.8$, $x_2 = -0.75$, what class?

- A. Class 0, decision value lower than 0
- B. Class 0, decision value higher than 0
- C. Class 1, decision value lower than 0
- D. Class 1, decision value higher than 0

Answer: A

Explanation

$0.43(0.8) + 2.55(-0.75) + 0.67 = -0.8985 < 0 \rightarrow \text{Class 0.}$

Theory

SVM:

Boundary: $w_1x_1 + w_2x_2 + b = 0$

$f(x) > 0 \rightarrow \text{Class 1}$

$f(x) < 0 \rightarrow \text{Class 0}$

Key concepts:

Support vectors: closest points

Margin: distance to support vectors

Kernel trick: higher dimensions

Exam tip: Positive = Class 1, negative = Class 0.

Question

Which model correctly identifies the highest percent of defaulted loans?

Model	Acc	Prec	Rec	F1
A (Logistic)	0.71	0.60	0.28	0.38
B (Tree)	0.66	0.43	0.17	0.24
C (SVM)	0.63	0.44	0.51	0.47
D (KNN)	0.68	0.50	0.15	0.23

- A. Model A
- B. Model B
- C. Model C
- D. Model D

Answer: C

Explanation

Model C has highest recall (0.51) = true positive rate.

Theory

Confusion matrix:

	Predicted +	Predicted -
Actual +	TP	FN
Actual -	FP	TN

Metrics:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total}$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1} = 2 \times \text{Prec} \times \text{Rec} / (\text{Prec} + \text{Rec})$$

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$$

Exam tip: “Percent of defaulted loans caught” = recall.

Question

What does tokenization do?

- A. Removes words with no value
- B. Separates text into words
- C. Converts words to numeric vectors
- D. Replaces words with similar meanings

Answer: B

Explanation

Tokenization = split text into words/tokens.

Question

What does CBoW represent?

- A. Recurrent neural network
- B. Fill-in-the-blank technique predicting a missing word
- C. Skip-gram
- D. Auto-encoder

Answer: B

Explanation

CBoW = fill-in-the-blank. Predicts missing word from context.

Theory

Word2Vec:

- Converts words to dense vectors
- Captures semantic relationships

CBoW:

- Predicts missing word from context
- Multiple inputs, one output

Skip-gram:

- Predicts context from target word
- One input, multiple outputs
- Better with rare words

Exam tip: CBoW = many in, one out. Skip-gram = one in, many out.

Question

What is an advantage of co-training?

- A. Different views of data; models learn from each other
- B. Higher productivity via parallel computing
- C. Improves accuracy by decreasing weights
- D. More reliable; data not interchangeable

Answer: A

Explanation

Co-training provides different views; models learn from each other.

Theory

Semi-supervised: small labeled + large unlabeled.

Techniques:

Self-training: predict unlabeled, add confident, repeat

Co-training: split features, two models label for each other

Unsupervised pre-processing: cluster then classify

Assumptions:

Smoothness: nearby points similar

Cluster: same cluster same label

Manifold: lower-dimensional

Question

What is the primary purpose of regularization?

- A. Increase model complexity
- B. Prevent overfitting by penalizing large coefficients
- C. Speed up training
- D. Increase features

Answer: B

Explanation

Regularization (Ridge, LASSO, Elastic Net) prevents overfitting.

Theory

Regularization:

Ridge (L2): $\lambda \sum \beta_j^2$; shrinks, keeps all

LASSO (L1): $\lambda \sum |\beta_j|$; can zero coefficients

Elastic Net: both

Exam tip:

Feature selection = LASSO

Shrink coefficients = Ridge

Both = Elastic Net

Question

What is the difference between Ridge and LASSO?

- A. Ridge L1; LASSO L2
- B. Ridge L2; LASSO L1
- C. Both L1
- D. Both L2

Answer: B

Explanation

Ridge = L2 (squared). LASSO = L1 (absolute), can zero coefficients.

Theory

Ridge (L2):

$$\text{Loss} = \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2$$

Shrinks toward zero, never exactly zero. Keeps all features.

LASSO (L1):

$$\text{Loss} = \text{RSS} + \lambda \sum_{j=1}^p |\beta_j|$$

Can zero coefficients. Feature selection.

Elastic Net: both penalties.

Question

What is the purpose of cross-validation?

- A. Increase training size
- B. Estimate model performance on unseen data
- C. Reduce features
- D. Speed up training

Answer: B

Explanation

Cross-validation estimates generalization to unseen data.

Theory

Cross-validation:

K-fold: train on $K-1$, validate on 1, repeat K times

Stratified: preserve class proportions

LOO: $K = N$

Time-series: respect temporal order

Benefits:

Reliable estimate

Uses all data

Detects overfitting

Exam tip: CV is for evaluation, not training.

Question

What is the bias-variance tradeoff?

- A. Accuracy vs training time
- B. Underfitting vs overfitting
- C. Features vs observations
- D. Precision vs recall

Answer: B

Explanation

High bias = underfitting. High variance = overfitting.

Theory

Total error = $\text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$.

Bias: overly simple assumptions (underfitting).

Variance: sensitivity to training data (overfitting).

Tradeoff:

Simple: high bias, low variance

Complex: low bias, high variance

Optimal: balance

Diagnose:

High train + high test error = underfitting

Low train + high test error = overfitting

Question

What is the purpose of a confusion matrix?

- A. Visualize decision boundary
- B. Summarize classification performance
- C. Plot ROC curve
- D. Cluster data

Answer: B

Explanation

Confusion matrix shows TP, TN, FP, FN.

Theory

Confusion matrix:

	Predicted +	Predicted -
Actual +	TP	FN
Actual -	FP	TN

Metrics:

$$\text{Accuracy} = (TP+TN)/(TP+TN+FP+FN)$$

$$\text{Precision} = TP/(TP+FP)$$

$$\text{Recall} = TP/(TP+FN)$$

$$\text{Specificity} = TN/(TN+FP)$$

$$F1 = 2 \times \text{Prec} \times \text{Rec} / (\text{Prec} + \text{Rec})$$

Question

What does the AUC-ROC curve measure?

- A. Regression accuracy
- B. Classifier's ability to distinguish classes across thresholds
- C. Clustering speed
- D. Number of clusters

Answer: B

Explanation

AUC-ROC measures discrimination across all thresholds.

Theory

ROC curve:

X-axis: $FPR = 1 - \text{Specificity}$

Y-axis: $TPR = \text{Sensitivity}$

Each point = different threshold

AUC:

Range 0 to 1

0.5 = random

1.0 = perfect

Higher = better

Exam tip: AUC measures discrimination, not accuracy.

Question

What is the difference between precision and recall?

- A. $\text{Precision} = \text{TP}/(\text{TP}+\text{FP})$; $\text{Recall} = \text{TP}/(\text{TP}+\text{FN})$
- B. $\text{Precision} = \text{TP}/(\text{TP}+\text{FN})$; $\text{Recall} = \text{TP}/(\text{TP}+\text{FP})$
- C. Same thing
- D. Precision for regression; recall for classification

Answer: A

Explanation

Precision = accuracy of positive predictions. Recall = capture of actual positives.

Theory

Precision: Of predicted positives, how many are correct?

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall: Of actual positives, how many caught?

$$\text{Recall} = \frac{TP}{TP + FN}$$

When:

High precision: false positives costly (spam)

High recall: false negatives costly (cancer, fraud)

F1: harmonic mean

Question

Rare disease (0.1%). Untreated = fatal. Which metric?

- A. Sensitivity
- B. Precision
- C. Accuracy
- D. Specificity

Answer: A

Explanation

Sensitivity (true positive rate) because missing cases is fatal.

Theory

Sensitivity:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

Use when false negatives costly (miss disease).

Specificity:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Use when false positives costly (unnecessary treatment).

Exam tip: Rare + fatal = sensitivity.

Question

10,000 transactions, 20 fraudulent. What bias?

- A. Measurement bias
- B. Representation bias
- C. Data composition bias
- D. Historical bias

Answer: C

Explanation

Imbalanced dataset = data composition bias.

Theory

Types:

Measurement: features measured differently

Representation: sample not representative

Data composition: imbalanced dataset

Historical: past discrimination encoded

Problem specification: wrong target

Deployment: wrong context

Exam tip: Imbalanced = data composition bias.

Question

Best approach to ensure education level does not affect predictions?

- A. Remove education level
- B. Replace with average during prediction
- C. Add interaction terms
- D. Create dummy variables

Answer: B

Explanation

Replacing with average ensures equal treatment.

Theory

Approaches:

Pre-processing: fix data

In-processing: modify algorithm

Post-processing: adjust predictions

Techniques:

Remove sensitive features (proxy effects remain)

Replace with average during prediction

Fairness constraints

Reweight samples

Question

How does demographic parity contribute to group fairness?

- A. Equal accuracy
- B. Identical predicted positive rates across subgroups
- C. Identical sensitivity
- D. Equal precision

Answer: B

Explanation

Demographic parity = same predicted positive rate across groups.

Theory

Group fairness:

Demographic parity: equal prediction rates

Equal opportunity: equal true positive rates

Equalized odds: equal TPR and FPR

Predictive parity: equal precision

Individual fairness: similar individuals get similar predictions.

Exam tip: Demographic parity = equal prediction rates.

Question

Chatbot gives incorrect info. Best action?

- A. Upgrade model
- B. Develop ethics framework
- C. Periodic risk reports
- D. Human verification mechanisms

Answer: D

Explanation

Human verification mitigates incorrect info risk.

Theory

Human-in-the-loop:

- Critical for high-stakes decisions
- Mitigates AI errors
- Required by many regulations

Applications:

- Chatbots: human review
- Medical: doctor confirms
- Credit: loan officer reviews
- Fraud: analyst investigates

Question

What is the difference between model risk and algorithmic bias?

- A. Model risk = errors; algorithmic bias = unfair discrimination
- B. Model risk = unfair discrimination; algorithmic bias = errors
- C. Same thing
- D. Model risk for AI; bias for statistics

Answer: A

Explanation

Model risk = errors/misuse. Algorithmic bias = unfair discrimination.

Theory

Model risk:

Adverse consequences from errors

Sources: assumptions, data, implementation, misuse

Applies to all models

Algorithmic bias:

Systematic unfair discrimination

Sources: biased data/features/objectives

Affects protected groups

Relationship: Bias is a source of model risk.

Question

What is the purpose of explainability?

- A. Increase accuracy
- B. Provide human-understandable reasons for decisions
- C. Reduce training time
- D. Increase features

Answer: B

Explanation

Explainability = human-understandable reasons for decisions.

Theory

Explainability:

Why did the model decide this?

Which features mattered?

How would predictions change?

Techniques:

SHAP: marginal contribution

LIME: local approximation

Feature importance: global ranking

Partial dependence: feature effect

Why: Compliance (GDPR), trust, validation.

Question

What is the difference between interpretability and explainability?

- A. Interpretability = internal mechanisms; explainability = decisions
- B. Interpretability = decisions; explainability = internal
- C. Same thing
- D. Interpretability for linear; explainability for NN

Answer: A

Explanation

Interpretability = understand internal workings. Explainability = explain decisions.

Theory

Interpretability:

Can you understand internal logic?

Linear regression: yes

Deep NN: no (black box)

Explainability:

Can you explain a decision?

Post-hoc for black-box

SHAP, LIME

Exam tip: Interpretability = internal. Explainability = external explanation.

Question

What is the “black box” problem?

- A. Models stored in black boxes
- B. Decisions cannot be understood by humans
- C. Too large for memory
- D. Trained on black market data

Answer: B

Explanation

Black box = decisions cannot be understood.

Theory

Black box:

- Millions of parameters
- No simple formula
- Cannot trace inputs to outputs

Consequences:

- Hard to validate
- Hard to audit
- Hard to trust
- Regulatory challenges

Solutions:

- Interpretable models
- Post-hoc explainability
- Attention mechanisms
- Hybrid models

Question

What is the primary concern with AI in hiring?

- A. Too slow
- B. May perpetuate historical bias
- C. Too expensive
- D. Cannot process resumes

Answer: B

Explanation

AI in hiring may perpetuate historical bias.

Theory

Risks:

- Historical bias
- Proxy discrimination
- Lack of transparency
- Regulatory risk

Mitigation:

- Audit training data
- Remove/de-bias proxies
- Fairness constraints
- Explanations to candidates
- Human review

Question

Which technique explains marginal effect of each feature?

- A. Shapley values
- B. Feature importance
- C. Grid search
- D. Feature scaling

Answer: A

Explanation

Shapley values = marginal contribution of each feature.

Theory

Shapley values:

- From game theory
- Fairly distributes prediction among features
- Considers all subsets

Properties:

- Additivity
- Consistency
- Symmetry

vs. Feature importance:

- Feature importance: global
- Shapley: per-prediction

Question

Customers concerned about loan decisions. Source of reputational risk and mitigation?

- A. Privacy; secure data
- B. Bias; diverse training
- C. Lack of explainability; provide reasons
- D. Lack of transparency; disclose data

Answer: C

Explanation

Lack of explainability; provide reasons for rejection.

Theory

Sources:

- Lack of explainability
- Algorithmic bias
- Privacy breaches
- Lack of transparency

Mitigation:

- Provide reasons
- Bias audits
- Secure data
- Disclose AI use

Question

Advantage of a practical ethics framework?

- A. Clear guidelines for ethical dilemmas
- B. Eliminates conflicts of interest
- C. Guarantees legal compliance
- D. Focuses on profit

Answer: A

Explanation

Provides clear guidelines for ethical dilemmas.

Theory

Practical ethics framework:

- Guidelines for dilemmas
- Consistency
- Accountability
- Builds trust

Does NOT:

- Eliminate conflicts
- Guarantee compliance
- Focus on profit
- Replace judgment

Components: Principles, processes, roles.

Question

Most likely advantage of AI ethics framework?

- A. Increases accuracy
- B. Strengthens firewall
- C. Increases testing productivity
- D. Reduces regulatory non-compliance risk

Answer: D

Explanation

Reduces regulatory non-compliance risk.

Theory

Why ethics frameworks matter:

- Regulatory compliance (EU AI Act, GDPR, CCPA)
- Reputation
- Trust
- Risk management

Key regulations:

- EU AI Act: risk-based
- GDPR: right to explanation
- CCPA: California privacy
- SEC/FCA: model risk

Question

Which is correct regarding Utilitarianism?

- A. Can lead to violations of individual rights
- B. Focuses on virtuous character
- C. Relies on laws
- D. Values intrinsic morality

Answer: A

Explanation

Utilitarianism maximizes overall good; can violate individual rights.

Theory

Three frameworks:

Consequentialism (Utilitarianism): judges by consequences; maximize good; can violate rights

Deontology: judges by rules/duties; inherent right/wrong

Virtue Ethics: character traits; what would a virtuous person do?

Exam tip: Utilitarianism = consequences. Deontology = rules. Virtue = character.

Question

Which best describes deontology?

- A. Judges by consequences
- B. Judges by adherence to duties and rules
- C. Single quantifiable metric
- D. Emphasizes virtuous character

Answer: B

Explanation

Deontology judges by duties and rules, not consequences.

Theory

Deontology:

- Actions right/wrong in themselves
- Consequences do not determine morality
- Categorical imperative (Kant)

vs Consequentialism:

- Consequentialism: ends justify means
- Deontology: means matter

AI: Certain uses inherently wrong (mass surveillance).

Question

Which is correct regarding justice in AI ethics?

- A. Conflicts exist between distributive justice concepts
- B. Impartial application not required for procedural justice
- C. Personal autonomy secondary to social justice
- D. Meritocracy not a distributive justice principle

Answer: A

Explanation

Different justice concepts (egalitarianism, utilitarianism) can conflict.

Theory

Types:

- Distributive: fair allocation

 - Egalitarianism: equal

 - Utilitarianism: maximize welfare

 - Meritocracy: based on merit

- Procedural: fair processes, impartial law

- Social: fairness in society

Conflicts: Concepts can conflict.

Question

AI recommends treatment conflicting with patient preferences. Which principle violated?

- A. Explainability
- B. Beneficence
- C. Autonomy
- D. Non-maleficence

Answer: C

Explanation

Autonomy = respect patient's right to decide.

Theory

Five principles:

Nonmaleficence: do no harm

Beneficence: do good

Justice: fairness

Autonomy: respect choice

Explainability: understandable reasons

Autonomy in AI:

Right to refuse treatment

Consent to AI decisions

AI supports, not replaces, judgment

Question

Model trained on US travelers, fails in Brazil/Nigeria/Thailand. Cause?

- A. Problem specification bias
- B. Deployment bias
- C. Historical bias
- D. Sampling bias

Answer: D

Explanation

Training sample not representative = sampling bias.

Theory

Sampling bias:

- Training from one population
- Deployed in different population
- Poor performance on new population

Mitigation:

- Representative data
- Diverse testing
- Monitor subgroups
- Transfer learning

Question

Primary purpose of AI ethics framework?

- A. Maximize accuracy
- B. Provide guidelines for ethical AI
- C. Reduce training time
- D. Increase features

Answer: B

Explanation

Provides guidelines for ethical AI development and use.

Theory

Framework:

- Defines principles
- Establishes processes
- Assigns roles
- Provides training

Benefits:

- Reduces regulatory risk
- Builds trust
- Consistency
- Protects reputation

Question

Crucial consideration when evaluating model landscape?

- A. Models operate independently
- B. All use same input data
- C. Consolidate models
- D. Interdependencies identified

Answer: D

Explanation

Model landscape shows interactions; identify interdependencies.

Theory

Model landscape:

- Shows relationships
- Identifies critical models
- Maps data flows
- Highlights interdependencies

Why:

- One output = another's input
- Failure cascades
- Correlated errors
- Systemic risk

Question

Which correctly describes roles?

- A. Board maintains inventory
- B. Model risk function sets appetite
- C. Model ownership never shared
- D. Auditors are independent checks

Answer: D

Explanation

Auditors check validation. Board sets appetite. Model risk function maintains inventory.

Theory

Roles:

Board: sets risk appetite

Model risk function: inventory, policies

Developers: build models

Validators: independent validation

Internal auditors: check validation

External auditors: external check

Model owners: accountable

Question

Which team ensures IT integration without disruption?

- A. Core Implementation
- B. System Implementation
- C. Model Design
- D. Data Management

Answer: B

Explanation

System Implementation Team integrates model with IT systems.

Theory

Teams:

Model Design: architecture, methodology

Core Implementation: builds/codes

System Implementation: IT integration

Data Management: data quality

System Implementation:

IT integration

Production function

Deployment

Monitoring

Question

New model outperforms traditional. Next step?

- A. Deploy to production
- B. Inform IT
- C. Retire existing model
- D. Request validation

Answer: D

Explanation

Validate before deployment. Never deploy without validation.

Theory

Validation:

- Independent team
- Conceptual soundness
- Data quality
- Performance testing
- Implementation review

When:

- Before deployment
- Periodically
- After changes
- On degradation

Question

AI credit model uses school transcripts, social media. Validation finding?

- A. No concerns
- B. Raise concerns about sensitive data
- C. Approve if backtest improves
- D. Add more alternative data

Answer: B

Explanation

Sensitive alternative data requires heightened scrutiny.

Theory

Alternative data:

- Social media, transcripts, rent
- Expands credit access
- Ethical/legal concerns

Governance:

- Provenance: legal right
- Consent: applicant approval
- Privacy: protect info
- Fairness: avoid discrimination
- Transparency: disclose use

Question

Advantage of increasing monitoring frequency?

- A. Reduces need for retraining
- B. Detects degradation or drift
- C. Minimizes IT involvement
- D. Improves interpretability

Answer: B

Explanation

Detects degradation/drift for timely intervention.

Theory

Monitoring:

- Detect performance degradation
- Identify data drift
- Identify concept drift
- Trigger retraining

Drift types:

- Data: input distribution changes
- Concept: input-output relationship changes
- Label: target distribution changes

Question

Purpose of a model inventory?

- A. Store code
- B. List all models and track risk
- C. Train models
- D. Visualize performance

Answer: B

Explanation

Lists all models with purpose, risk, owner, validation status.

Theory

Model inventory:

- Maintained by model risk function
- Tracks all models
- Documents purpose, owner, risk
- Records validation status
- Identifies interdependencies

Contents:

- Name, description
- Purpose
- Owner, developer
- Risk classification
- Validation status

Question

Purpose of model documentation?

- A. Increase accuracy
- B. Record design, assumptions, limitations
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Records design, assumptions, limitations for validation and audit.

Theory

Documentation:

- Purpose and use
- Methodology
- Data sources
- Limitations
- Performance
- Validation results
- Usage guidelines

Why: Validation, audit, knowledge transfer, compliance.

Question

Purpose of use test?

- A. Test production use
- B. Assess appropriateness for intended use
- C. Test speed
- D. Test accuracy

Answer: B

Explanation

Use test assesses fit for intended purpose.

Theory

Use test:

- Used as intended?
- Users understand limitations?
- Appropriate for context?
- Outputs interpreted correctly?

Questions:

- What decisions supported?
- Who uses it?
- Are assumptions understood?
- Alternatives?

Question

Difference between white-box and black-box testing?

- A. White-box = internal logic; black-box = inputs/outputs
- B. White-box = inputs/outputs; black-box = internal
- C. Same
- D. White-box for AI; black-box for statistics

Answer: A

Explanation

White-box = internal. Black-box = external.

Theory

White-box:

- Internal logic and code
- Model structure knowledge
- Component tests
- Implementation errors

Black-box:

- Inputs/outputs only
- No internal knowledge
- Overall behavior
- Real-world simulation

Both needed.

Question

Purpose of data provenance?

- A. Verify legal right to use data
- B. Increase volume
- C. Speed up training
- D. Reduce quality

Answer: A

Explanation

Provenance = legal right to use data.

Theory

Data provenance:

Where from?

Legal right?

Consent obtained?

Accurate?

Why:

Legal compliance (GDPR, CCPA)

Ethical use

Regulatory

Reputation

Question

Purpose of data classification?

- A. Increase volume
- B. Categorize by sensitivity and structure
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Categorizes by sensitivity (public, internal, confidential) and structure.

Theory

Classification:

Sensitivity:

Public, Internal, Confidential, Restricted

Structure:

Structured (tables), Unstructured (text, images)

Purpose: Determine handling, access, protection.

Question

Purpose of metadata management?

- A. Increase volume
- B. Document origin, structure, definitions, relationships
- C. Speed up training
- D. Reduce quality

Answer: B

Explanation

Metadata = data about data.

Theory

Metadata:

- Origin, history
- Structure, format
- Definitions, meaning
- Relationships
- Access permissions

Why:

- Discovery
- Lineage
- Compliance
- Governance

Question

Purpose of data access controls?

- A. Increase volume
- B. Only authorized users access data
- C. Speed up training
- D. Reduce quality

Answer: B

Explanation

Ensures only authorized users access data.

Theory

Access controls:

- RBAC: role-based
- ABAC: attribute-based
- Least privilege
- Need-to-know

Why:

- Privacy
- Security
- Compliance
- Prevent misuse

Question

Purpose of data governance framework?

- A. Increase volume
- B. Ensure quality, security, compliance
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Ensures data quality, security, compliance, proper use.

Theory

Data governance:

- Strategy and vision
- Quality
- Provenance
- Classification
- Metadata
- Protection
- Access
- Compliance
- Roles

Question

Purpose of model risk management?

- A. Increase accuracy
- B. Identify, assess, mitigate model risks
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Identifies, assesses, mitigates model risks.

Theory

Model risk management:

- Identification, inventory
- Risk assessment
- Development, documentation
- Independent validation
- Monitoring
- Governance

Sources:

- Wrong assumptions
- Bad data
- Implementation errors
- Misuse
- Drift

Question

Role of model risk function?

- A. Build models
- B. Maintain inventory and develop policies
- C. Use models
- D. Reduce features

Answer: B

Explanation

Maintains inventory, develops policies, oversees model risk.

Theory

Responsibilities:

- Maintain inventory
- Develop policies
- Classify risk
- Coordinate validation
- Monitor performance
- Report to management
- Ensure compliance

Independence: From developers.

Question

Purpose of periodic model validation?

- A. Increase accuracy
- B. Confirm model remains fit for purpose
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Confirms model remains fit for purpose over time.

Theory

Periodic validation:

- Assumptions valid
- Data appropriate
- Performance as expected
- Methodology current
- Aligned with purpose

Frequency:

- Based on risk ranking
- Higher risk = more frequent
- Annually or risk-based
- Triggered by changes

Question

Purpose of model implementation tasks?

- A. Design model
- B. Deploy to production and integrate IT
- C. Validate model
- D. Reduce features

Answer: B

Explanation

Deploy to production, integrate IT systems.

Theory

Implementation:

- IT integration
- Data pipeline
- Performance testing
- User training
- Documentation
- Monitoring setup

Challenges:

- Legacy systems
- Data quality
- Scalability
- Security

Question

Purpose of model adaptation tasks?

- A. Design model
- B. Update model for changing conditions
- C. Validate model
- D. Reduce features

Answer: B

Explanation

Updates model for changing conditions.

Theory

Adaptation:

- Retraining
- Recalibration
- Update assumptions
- Add features
- Regulatory response

Triggers:

- Degradation
- Data drift
- Concept drift
- Business changes
- Regulatory changes

Question

Purpose of model risk appetite?

- A. Increase accuracy
- B. Define acceptable model risk level
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Defines acceptable model risk, set by board.

Theory

Risk appetite:

- Set by board
- Defines tolerance
- Guides management
- Aligns with overall appetite

Considerations:

- Business strategy
- Regulatory
- Risk capacity
- Stakeholder expectations

Question

Purpose of model governance policies?

- A. Increase accuracy
- B. Establish rules for development, validation, use
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Establishes rules for consistency and accountability.

Theory

Policies:

- Development standards
- Validation requirements
- Documentation standards
- Approval processes
- Monitoring
- Roles

Purpose:

- Consistency
- Accountability
- Compliance
- Risk management

Question

Purpose of documentation standards?

- A. Increase accuracy
- B. Ensure consistent documentation
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Ensures consistent and comprehensive documentation.

Theory

Standards:

- Purpose
- Methodology
- Data sources
- Limitations
- Performance
- Validation
- Usage

Benefits:

- Consistency
- Knowledge transfer
- Audit readiness
- Compliance

Question

Purpose of model inventory and landscape?

- A. Increase accuracy
- B. Track all models and interdependencies
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Inventory = list. Landscape = map of interactions.

Theory

Inventory:

- List of all models
- Name, purpose, owner
- Risk level
- Validation status

Landscape:

- Visual interactions
- Data flows
- Critical models
- Interdependencies

Question

Purpose of model risk reporting?

- A. Increase accuracy
- B. Inform management about model risk
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Informs management about model risk.

Theory

Reporting:

- Inventory summary
- Validation status
- Key issues
- Performance
- Remediation

Audience:

- Board
- Senior management
- Risk committee
- Regulators

Question

Purpose of model change management?

- A. Increase accuracy
- B. Control changes with review
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Controls changes with review, approval, documentation.

Theory

Change management:

- Identify need
- Assess impact
- Obtain approval
- Implement
- Document
- Validate

Types:

- Minor: parameter updates
- Major: methodology changes
- Emergency: urgent fixes

Question

Purpose of model review?

- A. Increase accuracy
- B. Periodically assess performance and fitness
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Periodically assesses performance and fitness.

Theory

Review:

- Performance
- Assumptions
- Data quality
- Methodology
- Documentation

Outcomes:

- Continue
- Modify
- Replace
- Retire

Question

Purpose of model retirement?

- A. Increase accuracy
- B. Formally decommission unused model
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Formally decommissions unused models.

Theory

Retirement:

- Document reason
- Obtain approval
- Transition
- Archive
- Update inventory

Reasons:

- No longer needed
- Better model
- Regulatory changes
- Business changes

Question

Purpose of model risk culture?

- A. Increase accuracy
- B. Promote awareness and accountability
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Promotes awareness and accountability for model risk.

Theory

Culture:

- Awareness
- Accountability
- Open communication
- Continuous learning
- Ethical behavior

Building blocks:

- Tone from top
- Training
- Aligned incentives
- Transparency

Question

Purpose of model risk training?

- A. Increase accuracy
- B. Ensure staff understand model risk
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Ensures staff understand model risk and responsibilities.

Theory

Training:

- What is model risk?
- Roles and responsibilities
- Policies
- Best practices
- Case studies

Audience:

- Board
- Management
- Developers, validators
- Users, all staff

Question

Which statement correctly describes inscrutability?

- A. Cannot understand internal mechanisms
- B. Cannot determine accuracy
- C. Cannot explain predictions
- D. Cannot generalize

Answer: A

Explanation

Inscrutability = cannot understand internal mechanisms.

Theory

Inscrutability:

- Cannot understand internal workings
- Hard to validate, audit
- Regulatory challenges

Mitigation:

- Interpretable models
- Post-hoc explainability
- Documentation
- Human oversight

Question

Primary purpose of data governance framework?

- A. Increase volume
- B. Ensure quality, security, compliance
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Ensures quality, security, compliance.

Theory

Data governance:

- Strategy
- Quality
- Provenance
- Classification
- Metadata
- Protection
- Access
- Compliance
- Roles

Question

Purpose of model risk appetite?

- A. Increase accuracy
- B. Define acceptable risk level
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Defines acceptable model risk level.

Theory

Risk appetite:

- Set by board
- Defines tolerance
- Guides management
- Aligns with overall appetite

Question

Purpose of model validation?

- A. Increase accuracy
- B. Independently assess soundness and fitness
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Independently assesses model soundness and fitness.

Theory

Validation:

- Conceptual soundness
- Data quality
- Performance
- Implementation
- Documentation

When:

- Before deployment
- Periodically
- After changes

Question

Purpose of model monitoring?

- A. Increase accuracy
- B. Detect degradation and drift
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Detects degradation and drift.

Theory

Monitoring:

- Performance metrics
- Data drift
- Concept drift
- Alerts
- Reporting

Triggers:

- Below threshold
- Significant drift
- Regulatory changes
- Business changes

Question

Purpose of model documentation?

- A. Increase accuracy
- B. Record design, assumptions, limitations
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Records design, assumptions, limitations.

Theory

Documentation:

- Purpose
- Methodology
- Data sources
- Limitations
- Performance
- Validation
- Usage

Question

Purpose of model inventory?

- A. Increase accuracy
- B. List all models and track risk
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Lists all models with risk and validation status.

Theory

Inventory:

Name, description

Purpose, use case

Owner, developer

Risk classification

Validation status

Last review

Maintained by: Model risk function.

Question

Purpose of model landscape?

- A. Increase accuracy
- B. Show interactions and interdependencies
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Shows interactions and interdependencies.

Theory

Landscape:

- Relationships
- Critical models
- Data flows
- Interdependencies

Why:

- Cascading failure
- Correlated errors
- Systemic risk

Question

Purpose of model risk reporting?

- A. Increase accuracy
- B. Inform management about model risk
- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Informs management about model risk.

Theory

Reporting:

- Inventory summary

- Validation status

- Key issues

- Performance

- Remediation

Audience: Board, management, risk committee, regulators.

Question

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- C. Speed up training
- D. Reduce features

Answer: B

Explanation

Controls changes with review, approval, documentation.

Theory

Change management:

- Identify need
- Assess impact
- Approval
- Implement
- Document
- Validate

Types:

- Minor: parameters
- Major: methodology
- Emergency: urgent

Key Takeaways

Module 1: AI and Risk

Inscrutability vs explainability vs interpretability

Classical AI vs deep learning

Data types, scaling, PCA

Module 2: Tools and Techniques

K-means, hierarchical, DBSCAN

WCSS, BCSS, silhouette, scree

Random forest, SVM, NLP, regularization

Metrics: precision, recall, F1, AUC

Module 3: Risk and Risk Factors

Sensitivity vs specificity

Algorithmic bias sources

Fairness metrics

Module 4: Responsible and Ethical AI

Ethical frameworks: consequentialism, deontology, virtue

Principles: beneficence, nonmaleficence, justice, autonomy, explainability

Bias, fairness, privacy

Module 5: Data and AI Model Governance

Data governance, model governance

Validation, monitoring, inventory, landscape

Roles and responsibilities